The Tests Passed. The Grader Said No.

Checking the checker in AI security repair · exploratory local study

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RESEARCH QUESTION

Can we judge an AI's decisions when its checker gives misleading feedback?

BACKGROUND

AI coding agents revise work using feedback [1, 2]. A faulty checker can mislead evaluation.

METHODS

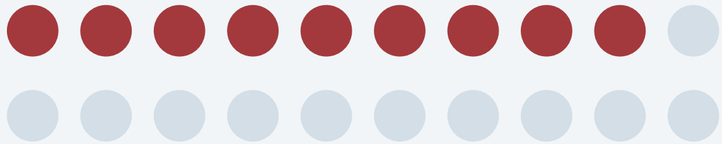
40 saved sessions · 1 local model
5 small Python security task types

1. **Read saved final test results**
2. **Compare with final verdicts**
3. **Inspect checker and source**

RESULTS · THE HIDDEN RULE

9 of 20

final submissions passed our tests but still received FAIL



9 passed tests · 11 failed tests
All 20 were rejected by our hidden rule.

Our rule was hidden from the AI.
Test pass ≠ proof of security.

RESULTS · STOPPING

2 min 8 sec

Median from stop acknowledgement to run end in 13 post-PASS sessions. A 14th timed out (excluded from median).

0 further logged repair attempts

CONCLUSION

Before judging the AI, check the grader.

We improved step logs and tested immediate stopping in the new collector.

NEXT STEPS

Independent task review → broader paired tests of true PASS vs. false FAIL.

REFERENCES & EVIDENCE

[1] Yao et al. ReAct (2023), arXiv:2210.03629. [2] Shinn et al. Reflexion (2023), arXiv:2303.11366.

[3] github.com/Imranqsl212/AiResearch · Evidence: paper/artifacts/results.json; analysis/HISTORICAL_TERMINATE_LATENCY_20261104.md